

New Attacks on QR-UOV Across Vinegar Dimensions: Direct Forgery and Equivalent-Key Recovery

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Abstract

Unbalanced Oil and Vinegar (UOV) offers short signatures and a simple signing procedure, but its public keys are large. QR-UOV reduces this cost by arranging the public quadratic map in blocks that represent elements of a quotient ring, chosen in the recommended parameters to be a degree-three extension field. NIST advanced QR-UOV to the third round of its Additional Digital Signatures process. Existing analyses cost direct forgery and structural key recovery as separate attacks at fixed parameter sets. This leaves open whether changing only the number of vinegar variables can make both attack paths harder while the output dimension remains fixed.

In this paper, we prove that it cannot, as measured by the square nonlinear system passed to the root solver. Let m be the expanded output dimension and let ℓ be the quotient degree. For every vinegar dimension V and every nonzero verification target, one of two attacks has a solver core of dimension at most $m - \ell - 1$. When $V \leq m - \ell - 1$, a regular solution of a V -variable directional system determines an equivalent signing key. When $V \geq m - \ell$, a direct attack zeros ℓ target coordinates, constructs a common isotropic space for the corresponding quadrics, and uses homogeneity to remove one more variable. For QR-UOV, $\ell = 3$, so the bound is $m - 4$ and the recommended parameter sets yield cores of dimensions 50, 74, and 101.

We estimate the cost of these cores with a finite-characteristic symbolic solver that uses local, rather than global, branch separation. In the inherited Boolean-gate model, the resulting costs are 140.384, 191.027, and 247.125 bits, below the corresponding category targets by 2.616, 15.973, and 24.875 bits. We also execute the complete attack pipeline on reduced parameters and obtain forgeries accepted by the unchanged public verifier. These experiments validate the algebraic composition at toy scale, not the full-parameter extrapolation. The concrete estimates remain conditional on the stated characteristic-polynomial degree transfer, the fast online-product schedule, the public-random expansion model, and astronomical symbolic memory.

1 Introduction

NIST opened its Additional Digital Signatures process to broaden the set of post-quantum signatures beyond the structured-lattice schemes selected in its first standardization process. Following the second round, NIST advanced nine candidates, including QR-UOV, to a third round of public analysis and permitted the teams to update their specifications and implementations [11, 12]. QR-UOV is attractive in this setting because it retains the simple signing procedure and short signatures of Unbalanced Oil and Vinegar (UOV) while addressing UOV’s main efficiency drawback: its large public key [1, 3].

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Oil and Vinegar signatures hide a linear subspace on which the central quadratic map becomes easy to invert. Kipnis and Shamir recovered this hidden structure in the balanced construction [5]. Kipnis, Patarin, and Goubin responded with UOV, which uses more vinegar variables than oil variables [4]. The vinegar-to-oil ratio is therefore a security parameter. Increasing it frustrates some key-recovery attacks, but it also gives a forger more freedom when solving the public equations. Braeken, Wolf, and Preneel made this second effect explicit by identifying highly unbalanced regimes that are vulnerable to direct algebraic attacks [6].

Concretely, UOV publishes a quadratic map from $v + m$ variables to m outputs over a finite field. A secret change of coordinates reveals v *vinegar variables* and m *oil variables*, with v of the former and m of the latter. The central equations contain no products of two oil variables. Once the vinegar variables are fixed, signing reduces to a linear system in the oil variables. The public key must nevertheless describe m dense quadratic forms, each with quadratically many coefficients. This cost motivated several structured variants, including field lifting and block-anti-circulant UOV [7, 8]. The latter was broken by decomposing the reducible quotient-ring structure behind its public blocks [9].

QR-UOV keeps the block-compression idea but represents the public matrices by elements of a quotient ring $\mathbb{F}_q[X]/(f)$ and chooses f to be irreducible [1, 3]. For the recommended parameters, $\deg f = \ell = 3$, so the quotient ring is the field $\mathbb{F}_{q^3}$. Irreducibility prevents the decomposition used against block-anti-circulant UOV, but it does not erase the extension-field description. An attacker can still view the structured public key as a smaller UOV-like system with V vinegar variables and m/ℓ oil variables over $\mathbb{F}_{q^\ell}$. In the notation used throughout this paper, the expanded base-field dimension is

$$n = \ell V + m.$$

This description is the common starting point for the QR-UOV pull-back, lifting, and rectangular-MinRank analyses [1, 18]. It also shows why changing V affects two different attack surfaces at once. It changes the underdetermination of the base-field verification equations and the dimension of the hidden extension-field system (Section 3).

Prior work follows the two attack surfaces separately. Direct attacks reduce an underdetermined multivariate quadratic system before applying a polynomial solver [38, 39, 37, 21]. Key-recovery attacks instead search for the hidden oil space through reconciliation, intersection, or MinRank equations [10, 17, 18, 19]. Pébereau showed that one suitable oil vector can suffice to recover an equivalent UOV key and stated the resulting parameter tradeoff directly: a larger variable-to-equation ratio helps against key recovery but hurts against forgery [22]. The original QR-UOV paper reflects the same division when it chooses the vinegar count mainly against pull-back and lifting reconciliation attacks, while choosing the output count against direct forgery [1].

The QR-UOV specification and the team’s latest security response cost these attack families at fixed parameter sets [3, 36]. What the prior analyses do not provide is a single bound covering every integer V while m and the quotient degree remain fixed. We give such a bound by connecting a directional equivalent-key system to a projective direct system. We measure both by the number of variables in the square nonlinear system passed to the root solver, which we call the *solver core*.

This brings us to the question that governs the paper:

Question. *Can QR-UOV raise the nonlinear solver cores of both attacks above $m - 4$ by changing only the vinegar count while keeping the output dimension and quotient degree fixed?*

1.1 Our results and technical overview

We answer this question negatively. Our main result is the following fixed-output algebraic ceiling:

For every vinegar count and every nonzero verification target, one of two complementary attacks on QR-UOV has solver core dimension at most $m - 4$.

The statement, formalized in [Theorem 4.10](#), concerns the algebraic bottleneck shared by the two attacks. It does not assert that their running time or memory usage depends only on this dimension.

Projective direct forgery. For a fixed nonzero target, a public output transformation makes three target coordinates zero. We use Chevalley–Warning to construct an $(m - 3)$ -dimensional subspace on which the corresponding three quadratic forms vanish. The remaining zero-target equations restrict to homogeneous quadrics, and quotienting their common scaling turns the residual problem into a square projective system of dimension $m - 4$. For the three recommended parameter sets, this gives solver cores of dimensions 50, 74, and 101 ([Theorems 4.1](#) and [4.9](#)). The construction uses public attack coins, and every returned vector is checked against the unchanged public verification map. A solver failure can therefore force a retry, but it cannot produce an accepted false signature ([Section 4](#)).

Directional equivalent-key recovery. In extension-field coordinates, the hidden oil space is the graph of a linear map G . Fixing a public direction c produces a system of m quadrics in V variables with the planted root Gc . If the derivative at a recovered root has full rank, public graph identities determine all of G . Exact pull-back and signing checks then certify an equivalent key that signs every target accepted by the verifier ([Theorem 7.8](#)). Pébereau established the equivalent-key endpoint from one suitable oil vector [22]; our contribution is a public polynomial-system mechanism for finding and certifying the required information in QR-UOV.

The fixed-output security scissors. The direct and directional routes respond oppositely to V . When $V \leq m - \ell - 1$, directional recovery already has a core of dimension at most $m - \ell - 1$. When $V \geq m - \ell$, target-orthogonal slicing with ℓ equations followed by projectivization gives a direct core of that same dimension. Since these cases cover every integer V , the combined algebraic core satisfies

$$r_{\text{attack}}(V) \leq m - \ell - 1.$$

For QR-UOV, $\ell = 3$, so the ceiling is $m - 4$ ([Theorem 4.10](#)). Increasing only V therefore cannot move the two attacks past their crossing point. A parameter revision must also increase m , or change the construction so that one of the reductions no longer applies.

Concrete estimates and executable evidence. To cost these cores, we adapt a nonsingular-root homotopy to characteristic 127, batch the dense quadratic lifting operations, choose sparse working fields, and replace global branch separation by a local separator for one eligible target branch. In the inherited Boolean-gate model, the resulting diagnostic costs are $2^{140.384}$, $2^{191.027}$, and $2^{247.125}$ gates at Levels I, III, and V, respectively. These rows lie 2.616, 15.973, and 24.875 bits below their category targets ([Table 6](#)). The Level-I row is a candidate model-based break, not a practical computation: it depends on the stated characteristic-polynomial degree transfer and fast online-product schedule, and the direct symbolic representation has prohibitive memory requirements ([Sections 8](#) and [10](#)).

The reduced-parameter implementation executes the complete local-separator pipeline from public inputs and returns signatures accepted by the unchanged verifier over two toy fields. This experiment checks that the algebraic components compose, but it does not benchmark or validate the full-parameter symbolic schedule ([Section 9](#)). The direct and directional reductions, the

fixed-output scissors, and verification of every returned signature or key are algebraic statements. The root-density arguments use the stated public-random or ideal-primitive models, while the finite gate rows remain conditional symbolic estimates. We keep these claim boundaries explicit throughout the paper.

Organization. Section 2 positions the two attacks against prior work. Section 3 connects the extension-field description to the official verification map. Section 4 proves the projective direct forgery and the fixed-output scissors, while Sections 5 to 7 develop and verify directional recovery. Section 8 gives the concrete solver refinements, and Sections 9 and 10 separate the evidence, assumptions, cost, memory, and parameter consequences.

2 Related work

From Oil and Vinegar to QR-UOV. Kipnis and Shamir broke the balanced Oil and Vinegar construction by recovering its hidden linear structure [5]. Kipnis, Patarin, and Goubin then introduced UOV with more vinegar variables [4]. Subsequent analysis showed that this imbalance must be chosen against both structural attacks and the increasingly underdetermined public system [6]. Efforts to reduce UOV’s public key include field lifting and block-anti-circulant matrices [7, 8]. The attack on block-anti-circulant UOV traces its weakness to a reducible quotient polynomial [9]. QR-UOV replaces that polynomial with an irreducible one and develops the resulting quotient-ring construction and its pull-back and lifting analyses [1]. We take this compressed construction as fixed and study what remains possible when only its vinegar dimension changes.

Direct attacks on underdetermined MQ. Thomae and Wolf reduce an underdetermined quadratic system to a smaller square system before invoking a generic solver [38]. Furue, Nakamura, and Takagi improve the range in which this reduction applies, and Hashimoto divides the residual equations into smaller blocks [39, 37]. Hashimoto’s method is the classical benchmark used by the QR-UOV team. Its January 2026 response gives costs of 158, 217, and 285 bits for the recommended parameter sets [36]. The resulting affine residual dimensions 51, 75, 102 are therefore prior art. The later “Just Guess” algorithm pushes more of the computation into exhaustive search. It improves the quantum estimates for QR-UOV, but it does not improve the team’s classical Hashimoto estimates [21, 36]. Our direct attack starts from the same affine dimensions, but full target normalization makes the residual equations homogeneous. Passing to projective space removes the common scale and gives the smaller square cores 50, 74, 101.

Recovering the hidden oil space. The Kipnis–Shamir attack searches directly for the secret linear structure [5]. Reconciliation expresses membership in the oil space as quadratic equations, while Beullens’ intersection attack uses the intersection of transformed oil spaces to reduce the search [10, 17]. Pébereau strengthens the endpoint: for the practical UOV parameters he analyzes, one suitable oil vector is enough to recover an equivalent key in polynomial time [22]. We do not claim this one-vector endpoint. Our contribution is a public QR-UOV mechanism for finding the required information. A chosen extension-field direction exposes a planted root Gc , the derivative test certifies when that root determines the hidden graph, and exact signing checks certify the recovered equivalent key (Section 5).

QR-UOV-specific key recovery. The original QR-UOV analysis applies reconciliation and intersection after pulling the block matrices back to the quotient field or lifting the public map to an extension field [1]. Furue and Ikematsu instead form a rectangular-MinRank instance from

the same extension-field structure [18]. Suzuki et al. extend the admissible target ranks and connect this method to the singular geometry used by other UOV attacks [19]. More recent symmetric-algebra and truncated-ring methods improve key recovery for some UOV parameter sets [42, 20]. For the recommended QR-UOV sets, the truncated-ring calculation gives no improvement because the characteristic 127 already exceeds the relevant solving degree, and the reported odd-characteristic symmetric-algebra costs do not beat reconciliation. These works sharpen individual key-recovery costs at fixed parameters. None gives a bound that joins key recovery to direct forgery for every vinegar dimension.

Parameter tradeoffs and common UOV frameworks. Pébereau states the qualitative tradeoff that motivates our theorem: increasing the variable-to-equation ratio strengthens UOV against key recovery and weakens it against forgery, while decreasing the ratio has the opposite effect [22]. The original QR-UOV paper makes the same point through its parameter selection, assigning the vinegar count to reconciliation and the output count to direct forgery [1]. Li, Guo, and Ding later describe QR-UOV, MAYO, and SNOVA through a common tensor-product framework and reduce QR-UOV key-recovery accounting to an extension-field UOV instance [2]. We do not claim the qualitative tradeoff or the extension-field view. Our result is the exact two-case statement that every integer V leaves either a direct-forgery core or an equivalent-key core of dimension at most $m - \ell - 1$, with the direct side reaching this bound only after projectivization (Theorem 4.10).

Polynomial-system solving. The arbitrary-core engine is the nonsingular-solutions homotopy of Safey El Din and Schost [34], building on symbolic homotopy and geometric-resolution methods [35, 15]. We replace the start-system and separator choices responsible for its published characteristic restriction. The appendices retain independent checks using the near-quadratic solver of van der Hoeven–Lecerf and a locally closed Kronecker algorithm [14, 16]. These methods cost the square cores produced by our reductions; they do not supply the QR-UOV reductions or the fixed-output theorem.

Chevalley–Warning. The direct reduction uses the classical fact that, when the sum of the degrees is smaller than the number of variables, the number of common zeros over a finite field is divisible by the characteristic [40, 41]. Applied to three quadrics in seven variables, it guarantees a nonzero extension vector. Exhaustive projective search makes the theorem constructive, so no unpriced existence oracle is hidden in the preprocessing. Our use of this classical theorem is therefore algorithmic rather than a new finite-field existence result.

3 From the expanded key to universal forgery

The direct attack works with the official base-field map, while equivalent-key recovery works first with its quotient-ring expansion. Equivalent-key recovery is complete only after the recovered structure gives a base-field decomposition that can invert every verification target. This section defines the expansion model used for average-case claims and gives the two public checks that turn a recovered graph into a universal forgery.

3.1 Coordinates, parameters, and expansion model

The Round-2 parameter sets use $q = 127$ and quotient degree $\ell = 3$. The official map has $v = \ell V$ vinegar variables, $m = \ell O$ oil variables, and m outputs over $\mathbb{F}_q$. Its quotient-ring expansion has

the same m outputs in $N = V + O$ variables over

$$K = \mathbb{F}_{q^\ell} = \mathbb{F}_{127^3}, \quad Q = |K| = 2,048,383, \quad \log_2 Q = 20.966054 \dots$$

The three parameter sets are as follows.

Level	v	m	V	O	N	$e = m - V$
I	156	54	52	18	70	2
III	228	78	76	26	102	2
V	306	105	102	35	137	3

We write $D = 2^V$ for the Bézout degree of a smooth square core and $L_s = \mathbb{F}_{Q^s}$ for the solver's working field. All concrete theorems below concern these three sets, for which $m < q$.

Write the expanded public tuple as

$$P = (p_1, \dots, p_m), \quad p_i(x) = x^\top P_i x, \quad P_i = \begin{pmatrix} A_i & E_i \\ E_i^\top & C_i \end{pmatrix}.$$

In normalized public coordinates, the hidden oil space is the graph

$$\mathcal{O} = \left\{ \begin{pmatrix} -Gc \\ c \end{pmatrix} : c \in K^O \right\}. \quad (1)$$

It is totally isotropic for every public form,

$$u^\top P_i w = 0 \quad (u, w \in \mathcal{O}, 1 \leq i \leq m), \quad (2)$$

exactly when

$$C_i = G^\top E_i + E_i^\top G - G^\top A_i G. \quad (3)$$

The corresponding secret central form is

$$F_i = \begin{pmatrix} A_i & B_i \\ B_i^\top & 0 \end{pmatrix}, \quad F_i(v, o) = v^\top A_i v + 2v^\top B_i o, \quad B_i = E_i - A_i G. \quad (4)$$

Definition 3.1 (Public-random expanded key). Sample the symmetric blocks $A_1, \dots, A_m$ independently and uniformly from $\text{Sym}_V(K)$. Conditional on $A = (A_i)_i$, sample the secret graph $G \in K^{V \times O}$ from any distribution. Conditional on (A, G) , sample $E_1, \dots, E_m$ independently and uniformly from $K^{V \times O}$, and set each C_i by (3). No independence between G and A is assumed.

Definition 3.2 (Independent-graph specialization). The independent-graph model is Definition 3.1 with G independent of A . The marginal distribution of G may otherwise be arbitrary.

Proposition 3.3 (Central blocks remain uniform). In Definition 3.1, the blocks $B_i = E_i - A_i G$ are independent and uniform in $K^{V \times O}$, and $B = (B_i)_i$ is independent of (A, G) . Hence the pairs $(A_i, B_i)_{1 \leq i \leq m}$ are independent and uniform on $\text{Sym}_V(K) \times K^{V \times O}$, although G may remain correlated with A .

Proof. For fixed (A, G) , each map $E_i \mapsto E_i - A_i G$ is a translation of $K^{V \times O}$. It therefore preserves the uniform law and makes that law independent of (A, G) . Independence of the A_i and E_i gives the stated product distribution. $\square$

The equivalent-key analyses use Definition 3.1; their bad-key events depend on the uniform central pairs and on fresh attack coins, not on independence between G and A . The direct reduction is key by key, but its average-case residual-root theorem conditions on an off-oil point and then requires G to be independent of the central coefficient pairs. It therefore uses Definition 3.2. The ideal-XOF and ideal-cipher transfers, and the gap for fixed SHAKE or AES, are stated in Section F. These models govern only the probability of finding a certificate. Every returned key and signature is checked against the unchanged public map.

3.2 From a recovered graph to a signing key

Proposition 3.4 (A verified graph gives an exact UOV decomposition). Let $G' \in K^{V \times O}$ satisfy every identity (3), and put

$$B'_i = E_i - A_i G', \quad F'_i = \begin{pmatrix} A_i & B'_i \\ B'^{\top}_i & 0 \end{pmatrix}, \quad S_{G'} = \begin{pmatrix} I_V & G' \\ 0 & I_O \end{pmatrix}.$$

Then $P_i = S_{G'}^{\top} F'_i S_{G'}$ for every i . The specification's invertible pull-back therefore gives matrices $\hat{P}_i = \hat{S}_{G'}^{\top} \hat{F}'_i \hat{S}_{G'}$ over $\mathbb{F}_q$ for the unchanged verification map [3, Sec. 4.1].

Proof. Direct block multiplication gives the extension-field factorization because the lower-right block is zero exactly when (3) holds. Applying the specification's inverse pull-back gives the base-field factorization. The full matrix calculation appears in Section E. $\square$

For an expanded base-field decomposition $\hat{P}_i = \hat{S}^{\top} \hat{F}_i \hat{S}$, write the central variables as $(a, o) \in \mathbb{F}_q^v \oplus \mathbb{F}_q^m$. Fixing a makes the central map linear in o because every oil-oil block is zero. Define this linear map by

$$\mathsf{L}_{\hat{F}}(a) \in \mathbb{F}_q^{m \times m}, \quad (\mathsf{L}_{\hat{F}}(a)o)_i := \hat{F}_i(a, o) - \hat{F}_i(a, 0). \quad (5)$$

Proposition 3.5 (Public signing witness). If $\det \mathsf{L}_{\hat{F}}(a) \neq 0$, then $(\hat{F}, \hat{S}, a)$ is a publicly checkable signing certificate. For every target $y \in \mathbb{F}_q^m$, compute

$$o = \mathsf{L}_{\hat{F}}(a)^{-1}(y - \hat{F}(a, 0)), \quad x = \hat{S}^{-1}(a, o).$$

Then $\hat{P}(x) = y$.

Proof. Substitution gives $\hat{F}(a, o) = y$ and then $\hat{P}(x) = \hat{F}(\hat{S}x) = y$. The forged signatures need not match the honest signing distribution. $\square$

Corollary 3.6 (Key-only universal chosen-message forgery). A verified certificate from Proposition 3.5 lets an attacker forge an accepted QR-UOV signature on every chosen message without a signing query. Given a message M , choose any allowed salt r , compute

$$\mu = \text{SHAKE256}(\text{seed}_{\text{pk}} \parallel \text{BytesToBits}(M), 512), \quad y = \text{Hash}(\mu, r),$$

invert y with Proposition 3.5, and output (r, x) .

Proof. The verifier accepts exactly when $\hat{P}(x) = \text{Hash}(\mu, r)$ [3, Secs. 4.2–4.3]. The certificate inverts every target under this unchanged map. It neither recovers nor uses the secret seed. $\square$

3.3 Finding an invertible oil system

Fix nonzero $\xi \in \mathbb{F}_q^\ell$. For each quotient-ring vinegar block j , let a_j place ξ in block j and zero elsewhere, and set

$$M_j := \mathsf{L}_{\hat{F}}(a_j) \in \mathbb{F}_q^{m \times m}.$$

Write

$$\pi_m := \prod_{k=1}^m (1 - q^{-k}), \quad \sigma_m := 1 - \pi_m.$$

Proposition 3.7 (Basis-first signing search). For the planted decomposition in Definition 3.1, the matrices $M_1, \dots, M_V$ are independent and uniform. The following searches then give public signing certificates.

1. Testing basis assignments in order uses fewer than $\pi_m^{-1} < 1.008$ tests on average. Testing all V fails with probability σ_m^V , whose base-two logarithms at Levels I, III, and V are -362.823 , -530.280 , and -711.692 .
2. Testing only the first 22, 31, and 40 assignments at the three levels fails with base-two log probabilities

$$-153.502, \quad -216.298, \quad -279.095. \quad (6)$$

3. Pairing the basis matrices as $(M_1, M_2), (M_3, M_4), \dots$ gives a deterministic fallback. Let $\mathcal{E}_{\text{sign}}$ be the event that every chosen pair span contains only singular matrices. Its probability satisfies

$$\Pr[\mathcal{E}_{\text{sign}}] \leq \eta_{\text{sign}}. \quad (7)$$

Its base-two logarithms at Levels I, III, and V are below

$$-544.820, \quad -796.276, \quad -1068.687.$$

Outside this event, testing every M_j and then $A + tB$ for each chosen pair and $t \in \mathbb{F}_q^\times$ succeeds after at most $V + \lfloor V/2 \rfloor (q - 1)$ tests.

For any exact candidate, define

$$\Delta(c_1, \dots, c_V) := \det \left(\sum_{j=1}^V c_j M_j \right).$$

If Δ is nonzero, a uniform scalar-block vinegar assignment succeeds with probability at least $1 - m/q$. The expected numbers of assignments are at most 1.740, 2.592, and 5.773 at the three levels. This last claim is key by key, and every successful determinant test is an exact public certificate even for a candidate that is not planted.

Proof. The specification's regular representation maps each independent uniform entry of the central block B_i bijectively to one row block of M_j . Thus the M_j are independent and uniform. The remaining statements follow from the exact rank distribution of a uniform matrix, a count of singular two-dimensional spans, and Schwartz–Zippel. The formulas and calculations appear in [Section E](#). $\square$

4 Direct forgery in $m - 4$ variables

The direct attack works with the official base-field verification map

$$\hat{P} : \mathbb{F}_q^n \longrightarrow \mathbb{F}_q^m, \quad n = v + m, \quad q = 127.$$

Given a nonzero target y , the attack constructs an $(m - 3)$ -dimensional public subspace on which three zero-target equations vanish identically. On this subspace, the remaining zero-target equations form a square projective system of dimension $m - 4$. The final nonzero-target equation only recovers the affine scale. The resulting solver cores are

$$50, \quad 74, \quad 101$$

for Levels I, III, and V.

Remark 4.1 (Relation to Hashimoto and the specification). The affine reduction to $m - 3$ variables is prior art. Hashimoto's underdetermined quadratic-system method and the QR-UOV specification already analyze these residual dimensions [37, 3]. Our additional reduction uses full target

normalization and solves the homogeneous zero-target equations modulo scaling. This removes one more variable, giving cores of dimension $m - 4$. We also give a finite Chevalley–Warning construction, a fixed-key seed bound, an explicit nonsingular-root probability for the degree-three pull-back, and a finite-characteristic solver cost.

The algebraic reduction below is key by key. Its probability calculation uses the independent-graph model of Definition 3.2: the hidden graph G is independent of the product-uniform central coefficient pairs (A_i, B_i) . This independence is needed only for the direct attack’s average-case root analysis; the directional results continue to allow arbitrary correlation between G and A . In the ideal-XOF and ideal-cipher models, official public and secret expansion have this independence whenever their independently sampled seeds differ. Section F adds the seed-collision and buffer terms explicitly. No such transfer is claimed for the fixed SHAKE or AES functions.

4.1 From the target to a projective system

If $y = 0$, then $x = 0$ is already a valid preimage, so only the nonzero case needs analysis. Let $y \in \mathbb{F}_q^m$ be nonzero. Choose a public matrix $T_y \in \text{GL}_m(\mathbb{F}_q)$ whose first $m - 1$ rows span $y^\perp$ and whose last row has inner product one with y . Then

$$T_y y = (0, \dots, 0, 1)^\top. \quad (8)$$

Write $Q_1, \dots, Q_m$ for the corresponding output recombinations of the public forms. Since T_y is fixed by y and invertible, product-uniform public forms remain product-uniform in the public-random expanded-key model.

Only Q_1, Q_2, Q_3 are used to construct the attack subspace. Choose a uniformly random seven-dimensional subspace $S_0 \subset E$, restrict these three forms to S_0 , and exhaustively test the points of $\mathbb{P}(S_0) \simeq \mathbb{P}^6(\mathbb{F}_q)$. Chevalley–Warning guarantees a nonzero common zero because three quadrics have total degree six in seven variables. Denote the first such point by w .

The hidden oil space is not known to the attacker, so the algorithm cannot test whether w is off oil. The random seed subspace makes this test unnecessary; the next proposition bounds the failure probability for every fixed oil space.

Proposition 4.2 (A random seed misses every fixed oil space). Fix an arbitrary m -dimensional subspace $\hat{\mathcal{O}} \subset \mathbb{F}_q^n$ and choose a uniform seven-dimensional subspace $S_0 \subset \mathbb{F}_q^n$. Then

$$\Pr[S_0 \cap \hat{\mathcal{O}} \neq \{0\}] \leq \epsilon_{\text{seed}} := \frac{q^m - 1}{q - 1} \frac{q^7 - 1}{q^n - 1}. \quad (9)$$

Consequently, except with probability at most ϵ_{seed} over public attack coins, every nonzero common zero found inside S_0 is off oil. For the three official profiles,

$$\log_2 \epsilon_{\text{seed}} \leq -1048.291, \quad -1551.477, \quad -2096.594. \quad (10)$$

The bound holds for every fixed hidden oil space; it does not average over key generation.

Proof. There are $(q^m - 1)/(q - 1)$ projective oil points. For any fixed projective point $[u]$, a uniform seven-dimensional subspace contains $[u]$ with probability $(q^7 - 1)/(q^n - 1)$. A union bound proves (9). The numerical values substitute the three official pairs (n, m) . $\square$

Constructing the isotropic subspace. For a quadratic form Q over a field of odd characteristic, write

$$B_Q(u, z) = \frac{Q(u + z) - Q(u) - Q(z)}{2}$$

for its polar bilinear form. A subspace is *totally isotropic* for Q when Q vanishes identically on it.

Theorem 4.3 (Building a common isotropic subspace). Let $Q_1, \dots, Q_s$ be homogeneous quadrics on $\mathbb{F}_q^n$, and let $0 \neq w$ satisfy $Q_i(w) = 0$ for every i . If

$$n \geq (s+1)r + s, \quad (11)$$

then a deterministic finite algorithm constructs an r -dimensional subspace U containing w on which every Q_i vanishes identically.

At each extension step the algorithm uses linear algebra and examines at most

$$|\mathbb{P}^{2s}(\mathbb{F}_q)| = \frac{q^{2s+1} - 1}{q - 1} \quad (12)$$

projective points.

Proof. At each step, intersect the polar-orthogonal spaces of the current isotropic subspace. Condition (11) leaves at least $2s + 1$ quotient dimensions, where Chevalley–Warning supplies a nonzero common zero of the induced s quadrics [40, 41]. Lifting this zero extends the isotropic subspace by one dimension. The complete dimension count appears in Section D. $\square$

Corollary 4.4 (Affine reduction after target normalization). Let $P : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$ be any homogeneous quadratic map, let $y \neq 0$, and choose $1 \leq s < m$. After an invertible output recombination sends y to the last coordinate, a nonzero common zero of any s of the first $m - 1$ transformed quadrics can be extended to an $(m - s)$ -dimensional common totally isotropic subspace whenever

$$n \geq (s+1)m - s^2. \quad (13)$$

On that subspace, the affine problem has $m - s$ equations in $m - s$ variables.

Proof. Apply Theorem 4.3 with $r = m - s$. Its condition becomes $n \geq (s+1)(m - s) + s = (s+1)m - s^2$. The selected s forms vanish identically on the constructed subspace and have zero target, leaving the stated affine residual. $\square$

The affine formulation leaves one variable that homogeneity makes redundant. The next corollary removes it exactly.

Corollary 4.5 (Projective reduction after target normalization). Assume that q is odd. Under the hypotheses of Corollary 4.4, put $r = m - s$ and let U be the constructed r -dimensional subspace. Normalize the target to $(0, \dots, 0, 1)$ and write

$$H_1, \dots, H_{r-1}, H_0$$

for the restrictions to U of the remaining transformed forms, with H_0 corresponding to the nonzero target coordinate. Then:

1. affine solutions of

$$H_1(z) = \dots = H_{r-1}(z) = 0, \quad H_0(z) = 1 \quad (14)$$

modulo the pair $z \sim -z$ are in bijection with projective points $[z] \in \mathbb{P}(U)$ satisfying

$$H_1(z) = \dots = H_{r-1}(z) = 0 \quad (15)$$

and $H_0(z)$ a nonzero square in $\mathbb{F}_q$;

2. a solution of (14) has invertible affine Jacobian if and only if its projective point is nonsingular for (15); and

3. after choosing a uniform nonzero linear form $L \in U^*$, the chart $L(z) = 1$ turns (15) into $r - 1$ quadrics in $r - 1$ affine variables. Any fixed projective root lies in this chart with probability

$$1 - \frac{q^{r-1} - 1}{q^r - 1} > 1 - \frac{1}{q}.$$

Thus the square solver core has dimension $m - s - 1$ and geometric degree at most 2^{m-s-1} .

Proof. Homogeneity multiplies every quadratic value by λ^2 under $z \mapsto \lambda z$. Thus H_0 can be scaled to one exactly when its value is a nonzero square. Euler's identity identifies affine nonsingularity with projective nonsingularity after removing the radial direction. Counting the linear forms that vanish at a fixed projective point gives the chart probability. The full Jacobian argument appears in [Section D](#). $\square$

For the QR-UOV parameters, set $s = 3$ and $r = m - 3$. The subspace condition is

$$n \geq 4m - 9. \quad (16)$$

The three official profiles give

$$(n, 4m - 9) = (210, 207), \quad (306, 303), \quad (411, 411).$$

The resulting projective solver dimensions are

$$k = r - 1 = m - 4 \in \{50, 74, 101\}. \quad (17)$$

The seed root and every extension step use finite exhaustive search, so this reduction needs no existence oracle. Even after overcharging the required linear algebra and search, preprocessing costs at most $2^{64.501}$ Boolean gates across the three profiles. The complete ledger appears in [Section D](#).

4.2 Why a usable projective root exists

The degree-three pull-back does not produce every UOV quadric over $\mathbb{F}_q$. The root count therefore uses the exact local randomness that the pull-back does provide. Let $E = K^{V+O}$, viewed over $\mathbb{F}_q$, and let $\hat{O} \subset E$ be the hidden base-field oil space. In central coordinates, each public-random pulled-back form has the form

$$Q(v, o) = \text{Tr}_{K/\mathbb{F}_q} \left(\gamma(v^\top A v + 2v^\top B o) \right) \quad (18)$$

for fixed nonzero $\gamma \in K$ and uniform $A \in \text{Sym}_V(K)$ and $B \in K^{V \times O}$.

Lemma 4.6 (Evaluation and first-derivative randomness). Let $x \in E \setminus \hat{O}$, and let $U \subseteq E$ contain x . For a uniform form Q from (18), $Q(x)$ is uniform in $\mathbb{F}_q$. Conditioned on $Q(x) = a$, the derivative $dQ_x|_U$ is uniform among the linear forms $\lambda \in U^*$ satisfying $\lambda(x) = 2a$. Moreover, evaluation at two off-oil points is linearly dependent only when the points differ by a nonzero scalar in $\mathbb{F}_q$.

Proof. Nondegeneracy of the trace pairing lets the central bilinear form prescribe every value and derivative allowed by Euler's identity. Comparing the rank-one coefficient matrices at two points gives the last claim. The full surjectivity and degree-three descent argument appears in [Section D](#). $\square$

Let U be the subspace from [Theorem 4.3](#), and choose an $\mathbb{F}_q$ -basis to identify U with $\mathbb{F}_q^r$. Under the normalization (8), write

$$H_j = Q_{3+j}|_U \quad (1 \leq j \leq r-1), \quad H_0 = Q_m|_U.$$

The affine residual is

$$H_1(z) = \cdots = H_{r-1}(z) = 0, \quad H_0(z) = 1, \quad r = m - 3. \quad (19)$$

The projective solver uses only the first $r-1$ equations; H_0 filters and scales their projective roots.

The next statement gives an explicit success probability rather than assuming that a random-looking square system has a root. Let

$$\pi_{r-1}(q) = \prod_{j=1}^{r-1} (1 - q^{-j}). \quad (20)$$

Proposition 4.7 (Affine nonsingular-root probability). Condition on the selected forms, the attack coins used to construct U , and an off-oil seed point. Put

$$d = \dim_{\mathbb{F}_q}(U \cap \hat{O}), \quad \mu = 1 - q^{d-r}.$$

Then $d \leq r-1$. Over the independent remaining public-random QR-UOV forms, the probability that (19) has an $\mathbb{F}_q$ -rational root with invertible $r \times r$ Jacobian is at least

$$\frac{\mu \pi_{r-1}(q)^2}{2 + \mu - (q-1)q^{-r}}. \quad (21)$$

For $q = 127$ and each of $r = 51, 75, 102$, this is at least

$$p_* = 0.326336998767509 \dots \quad (22)$$

Proof. The off-oil seed point gives $d \leq r-1$. Let M count all off-oil roots of (19). Lemma 4.6 gives $\mathbb{E}[M] = \mu$. Pairing proportional and nonproportional points gives

$$\mathbb{E}[M^2] = 2\mu + \mu^2 - (q-1)\mu q^{-r}. \quad (23)$$

At a fixed root, the same lemma makes the Jacobian invertible with probability $\pi_{r-1}(q)$. If N counts these roots, then $\mathbb{E}[N] = \mu \pi_{r-1}(q)$ and $N^2 \leq M^2$. The second-moment inequality now gives (21); using $\mu \geq 1 - q^{-1}$ gives (22). The complete pair count appears in [Section D](#). $\square$

Corollary 4.8 (Probability of a usable projective root). Under the same conditioning, the probability that the projective system

$$H_1 = \cdots = H_{r-1} = 0 \quad \text{in } \mathbb{P}^{r-1}(\mathbb{F}_q)$$

has a nonsingular point $[z]$ for which $H_0(z)$ is a nonzero square is at least the bound in (21), and hence at least p_* for every official profile.

Proof. By [Corollary 4.5](#), every affine nonsingular root of (19) maps to a nonsingular projective root with the required nonzero-square value; the two affine roots z and $-z$ map to the same projective point. Existence of an affine root therefore implies the displayed projective event. $\square$

The factor $1/p_* = 3.0643169600 \dots$ costs 1.615566 bits in the work-to-success ratio. A random chart adds less than $\log_2(127/126) = 0.011405$ bits. Unlike a Poisson heuristic, the root bound follows from the QR-UOV pull-back distribution and a second-moment argument. The seed-disjointness statement is fixed-key; the residual-root statement averages over the independent-graph idealized expansion model. We do not claim that repeated invocations on every fixed seeded key are independent.

4.3 The verified attack

Theorem 4.9 (Verified QR-UOV direct forgery). Work in the independent-graph public-random model and the official degree-three pull-back. For any nonzero target y , the following public algorithm has one-sided error.

1. Normalize the target to $(0, \dots, 0, 1)$ as in (8).
2. Choose a uniform seven-dimensional seed subspace, enumerate its projective points, and take a nonzero common zero of the first three forms.
3. Use [Theorem 4.3](#) to construct an $(m - 3)$ -dimensional common totally isotropic subspace for those forms.
4. Choose a uniform nonzero chart form L on that subspace. Dehomogenize the $m - 4$ remaining zero-target quadrics in the chart $L = 1$ and apply [Theorem 7.6](#) to the resulting square system in

$$k = m - 4$$

variables.

5. For each represented $\mathbb{F}_q$ -rational root, recover its projective point, test whether the last quadratic value is a nonzero square, scale to make that value one, reverse the public transformations, and evaluate the unchanged public map.

Every returned vector is a valid signature preimage. Over the independent-graph public-random key and the public attack coins, one complete invocation reaches an off-oil residual with a represented nonsingular $\mathbb{F}_q$ -rational projective root whose last quadratic value is a nonzero square with probability at least

$$(1 - \epsilon_{\text{seed}})p_* \left(1 - \frac{1}{q}\right) \left(1 - \frac{2^{2k}}{|k'|}\right), \quad k = m - 4, \quad (24)$$

where $k'/\mathbb{F}_q$ is the solver working extension. Taking

$$[k' : \mathbb{F}_q] = 15, \quad 22, \quad 30$$

at Levels I, III, and V gives the finite costs reported in [Section 10](#). The inverse of (24) is used as a conservative work-to-success normalization; it is not asserted to be a fixed-key Las Vegas restart factor.

Proof. The first $m - 1$ target coordinates are zero. By [Proposition 4.2](#), the seed subspace is disjoint from the hidden oil space with probability at least $1 - \epsilon_{\text{seed}}$ for every fixed key. Chevalley–Warning then supplies an off-oil seed root, and [Theorem 4.3](#) constructs the advertised subspace. Conditional on the selected first three forms, the graph, the attack coins, and this off-oil seed point, the remaining transformed forms are still independent uniform pull-back forms. [Corollary 4.8](#) supplies a usable nonsingular projective root with probability at least p_* . A random nonzero chart contains a fixed such root with probability greater than $1 - 1/q$. The finite-characteristic solver includes every root in the chart with the separator probability in (24). Square testing and scaling recover the associated affine solutions exactly. Re-evaluation of all m original public forms rejects every malformed or extraneous candidate, so the algorithm cannot output an invalid signature. $\square$

4.4 Why changing only the vinegar count cannot help

Combining the direct reduction with directional equivalent-key recovery gives a bound for every vinegar dimension. Let the quotient degree be ℓ , so that

$$n = \ell V + m.$$

Applying Corollary 4.4 with a general value of s gives the required $(m - s)$ -dimensional common-isotropic subspace whenever

$$\ell V \geq s(m - s). \quad (25)$$

By Corollary 4.5, the corresponding nonzero-target solver core has dimension $m - s - 1$.

Theorem 4.10 (Fixed-output bound across all vinegar dimensions). Work over a field of odd characteristic. Fix m and quotient degree ℓ . For every nonzero target, within the two attack families analyzed in this paper an algebraic core of dimension at most $m - \ell - 1$ is available for every integer vinegar count V :

$$r_{\text{attack}}(V) \leq m - \ell - 1. \quad (26)$$

More precisely:

1. if $V \leq m - \ell - 1$, directional equivalent-key recovery uses a square core of dimension V ;
2. if $V \geq m - \ell$, the direct reduction with $s = \ell$, followed by projectivization, uses a square core of dimension $m - \ell - 1$.

These cases cover every integer V . For QR-UOV, $\ell = 3$, so the ceiling is $m - 4$.

Proof. In the first case, $V < m$, so the m directional equations admit a V -equation square recombination and the planted-root attack of Section 5 has core $V \leq m - \ell - 1$. In the second case,

$$\ell V \geq \ell(m - \ell),$$

which is (25) with $s = \ell$. The affine direct residual has dimension $m - \ell$ and Corollary 4.5 removes its radial coordinate, leaving core $m - \ell - 1$. Since consecutive integers $m - \ell - 1$ and $m - \ell$ meet the two inequalities, no case is omitted. $\square$

The theorem is an algebraic-core statement, not a claim that running time depends only on the number of variables. For the official degree-three sets, Proposition 4.2 and Corollary 4.8 supply the success factors charged in the concrete ledger. Its design implication is exact: once V reaches $m - \ell$, increasing V while holding m fixed no longer enlarges the projective direct-forgery core. Conversely, decreasing V into the other case makes directional recovery no larger than the same ceiling. Moving both attack lines therefore requires changing m as well as V , or modifying the construction so that one reduction no longer applies.

The direct construction and the complete local-separator pipeline have both been executed on reduced parameters. The exact traces and public replay checks appear in Sections D and I; these are correctness tests, not full-parameter benchmarks.

5 Equivalent-key recovery from a planted root

Fixing a public direction produces a quadratic system in only V variables. The hidden graph gives this system a planted root. If the derivative at that root has full rank, public linear algebra recovers every column of the graph. The attack then verifies the graph identities, the pull-back, the public-map factorization, and a signing witness. A separate projective route recovers the same oil space directly from the common derivative kernel.

5.1 A public system with a planted root

Fix $c \in K^O$. Substitute the public oil coordinate c and the negated vinegar coordinate $-x$ into each public form:

$$q_{i,c}(x) := P_i(-x, c) = x^\top A_i x - 2x^\top E_i c + c^\top C_i c, \quad x \in K^V. \quad (27)$$

The equations $q_{1,c} = \dots = q_{m,c} = 0$ are public. Define their public derivative matrix by

$$\mathcal{D}_c(x) := \begin{pmatrix} (E_1 c - A_1 x)^\top \\ \vdots \\ (E_m c - A_m x)^\top \end{pmatrix}. \quad (28)$$

Recall that $B_i = E_i - A_i G$, and let $\mathcal{B}(c)$ have i th row $(B_i c)^\top$.

Proposition 5.1 (A planted root for every direction). For every $c \in K^O$, the point $x_c = Gc$ is a common zero of (27). More precisely, for every $z \in K^V$,

$$q_{i,c}(Gc + z) = z^\top A_i z - 2(B_i c)^\top z. \quad (29)$$

Consequently,

$$\mathcal{D}_c(x_c) = \mathcal{B}(c), \quad J_q(x_c) = -2\mathcal{B}(c).$$

Proof. Expand (27) at $x = Gc + z$. The graph identity (3) makes the constant term zero, while the linear term is $-2(B_i c)^\top z$. This proves (29); differentiation gives the two matrix identities. $\square$

5.2 A root with full-rank derivative determines the graph

Proposition 5.2 (Exact graph completion). Let x be a common zero of all equations $q_{i,c}$ and suppose $\text{rank } \mathcal{D}_c(x) = V$. For $d \in K^O$, define

$$b_{c,d}(x)_i := c^\top C_i d - x^\top E_i d \quad (1 \leq i \leq m). \quad (30)$$

Then the public system

$$\mathcal{D}_c(x)y = b_{c,d}(x) \quad (31)$$

has at most one solution $y \in K^V$. If $x = Gc$, its solution is $y = Gd$. Solving (31) for the O coordinate vectors $d = e_j$ therefore recovers G exactly. Checking every identity (3) then certifies a valid expanded UOV decomposition.

Proof. Full column rank gives uniqueness. For $x = Gc$, multiply (3) by $c^\top$ on the left and d on the right. The result is

$$(E_i c - A_i Gc)^\top Gd = c^\top C_i d - (Gc)^\top E_i d,$$

which is row i of (31). Hence Gd is the unique solution. The verified-key conclusion follows from Theorem 3.4. $\square$

Remark 5.3 (A compact attack certificate). A successful branch is certified by (c, x, G) and one full-rank minor of $\mathcal{D}_c(x)$. A verifier checks the directional equations, the graph completion systems, all graph identities, the inverse pull-back, and a signing witness. The certificate makes no claim about other roots or solver completeness.

5.3 How often the derivative has full rank

Let

$$\rho_{m,V}(Q) := \prod_{a=0}^{V-1} (1 - Q^{a-m}), \quad \tau_{m,V}(Q) := 1 - \rho_{m,V}(Q). \quad (32)$$

Thus $\rho_{m,V}(Q)$ is the probability that a uniform $m \times V$ matrix over K has full column rank.

Proposition 5.4 (Rank of the planted derivative). For a fixed expanded key, suppose some maximal minor of $\mathcal{B}(c)$ is a nonzero polynomial in c . A fresh uniform $c \leftarrow K^O$ then satisfies

$$\Pr[\text{rank } \mathcal{B}(c) = V] \geq 1 - \frac{V}{Q}. \quad (33)$$

In the public-random model, $\mathcal{B}(c)$ is a uniform $m \times V$ matrix for every fixed nonzero c . Moreover, the coordinate-direction matrices $\mathcal{B}(e_1), \dots, \mathcal{B}(e_O)$ are independent, so

$$\Pr[\text{rank } \mathcal{B}(e_j) < V \text{ for every } j] = \tau_{m,V}(Q)^O. \quad (34)$$

Proof. Every maximal minor of $\mathcal{B}(c)$ is homogeneous of degree V , so Schwartz–Zippel gives (33). Under Definition 3.1, each B_i is an independent uniform linear map. This gives the fixed-direction law and independence across coordinate directions. The exact rank calculations appear in Section C. $\square$

The verified attack later uses a fresh random linear recombination to select V of the m equations. The supplement also gives a deterministic public list of only $V(m - V) + 1$ recombinations, together with exact list sizes and failure probabilities for the official parameters (Section C).

5.4 Recovering the oil space by linear algebra

The faster projective route from Section A instead finds a public common zero x on the oil space. Form

$$C_x = (P_1 x \ \dots \ P_m x) \in L^{(V+O) \times m}.$$

Lemma 5.5 (The common derivative kernel). If $x \in \mathbb{P}(\mathcal{O}_L)$, then $\mathcal{O}_L \subseteq \ker C_x^\top$. If $\text{rank } C_x = V$, equality holds.

Proof. For $u \in \mathcal{O}_L$, (2) gives $u^\top P_i x = 0$ for every i . If $\text{rank } C_x = V$, the kernel of its transpose has dimension $(V + O) - V = O$. $\square$

At the planted oil point, smoothness of the restricted core gives derivative rank V . The lemma therefore recovers the full oil space. Row reduction, descent to K , and the checks from Theorems 3.4 and 3.7 then produce a verified signing key. The attack does not need the oil space to be unique: it keeps every exactly verified candidate with a signing witness. Section G separately bounds the probability of a second K -rational oil space.

6 The solver needs only a full-rank Jacobian

Fix $c \in K^O$ and a public recombination $R \in K^{V \times m}$. Let

$$f = (f_1, \dots, f_V)^\top := R(q_{1,c}, \dots, q_{m,c})^\top$$

and put

$$h(x) := \det J_f(x). \quad (35)$$

The root solver used in the next section asks that the equations form a regular sequence and that every prefix ideal be radical, but only after localization. Taking h as the localization polynomial removes every point where the Jacobian is singular.

Lemma 6.1 (Localization gives a reduced regular sequence). Let k be a perfect field, let $f_1, \dots, f_V \in k[x_1, \dots, x_V]$, and let $h = \det J_f$. In the localization

$$S_h := k[x_1, \dots, x_V]_h,$$

the sequence $f_1, \dots, f_V$ is regular. For every $1 \leq s \leq V$, the ideal $(f_1, \dots, f_s)S_h$ is radical.

The proof appears at the start of the finite-field solver appendix in the technical supplement. It applies the Jacobian criterion to each prefix and then uses the complete-intersection criterion in the regular ring S_h .

If $\mathcal{B}(c)$ has full column rank and $R\mathcal{B}(c)$ is invertible, Proposition 5.1 gives

$$h(Gc) = (-2)^V \det(R\mathcal{B}(c)) \neq 0. \quad (36)$$

Corollary 6.2 (Only the planted simple root is needed). If $\mathcal{B}(c)$ has full column rank and $R\mathcal{B}(c)$ is invertible, then Gc belongs to the locally closed set

$$\{f_1 = \dots = f_V = 0, h \neq 0\}.$$

The input tuple $(f_1, \dots, f_V, h)$ satisfies hypotheses (H1)–(H2) of the locally closed Kronecker algorithm of Giménez et al. [16]. These conclusions are independent of every root and irreducible component contained in $h = 0$.

Proof. Equation (36) puts the planted root in the open set. Lemma 6.1 gives exactly the two algebraic conditions required after localization. $\square$

Thus this recovery route needs local smoothness at the planted root, not global smoothness of the projective core. Singular roots and components contained in $h = 0$ do not affect the conclusion.

7 Finite-characteristic nonsingular-root recovery

The directional reduction does not need a description of every component of the square core. It needs one isolated root at which the Jacobian is invertible. This section adapts the symbolic-homotopy algorithm of Safey El Din and Schost to that exact task in characteristic 127.

7.1 The solver target and its degree parameters

Let k be a perfect field and let

$$f = (f_1, \dots, f_V) \in k[X_1, \dots, X_V]^V$$

be a square system of affine quadrics. Write

$$Z_{\text{ns}}(f) := \{x \in \bar{k}^V : f(x) = 0, \det J_f(x) \neq 0\}. \quad (37)$$

Every point in $Z_{\text{ns}}(f)$ is isolated. Bézout's theorem gives

$$C := 2^V, \quad C_0 := 2^V + V2^{V-1} = 2^{V-1}(V+2). \quad (38)$$

Here C bounds the number of isolated roots, while C_0 bounds the degree of the one-parameter homotopy curve. Rational reconstruction of a degree- C_0 curve uses coefficients through precision $2C_0$; the lifting ledger below therefore runs to that precision. The second formula is the coefficient sum of $(\vartheta_0 + 2\vartheta)^V$ after discarding ϑ_0^2 and ϑ^{V+1} , exactly as in the multihomogeneous degree definition of Safey El Din and Schost [34].

The published algorithm starts from a product system with known simple roots, follows its C branches symbolically, specializes at the target fiber, and removes singular outputs by a Jacobian-based CLEAN operation. Its large-characteristic hypothesis supports the particular finite families used for the start and the separator. We replace both families.

7.2 A simple product start over any sufficiently large finite field

Lemma 7.1 (Vandermonde product start). Let k' contain $2V$ distinct elements $a_1, \dots, a_{2V}$. For $a \in k'$, put

$$L_a(X) := X_1 + aX_2 + \dots + a^{V-1}X_V + a^V, \quad (39)$$

and, for $1 \leq i \leq V$, set

$$g_i(X) := L_{a_{2i-1}}(X)L_{a_{2i}}(X). \quad (40)$$

Then $g = (g_1, \dots, g_V)$ has exactly $C = 2^V$ affine roots over k' . Every root is simple. A zero-dimensional parametrization of all roots can be constructed in $O(V^2C)$ operations in k' .

Proof. Choose one node $b_i \in \{a_{2i-1}, a_{2i}\}$ from each pair. The equations $L_{b_1} = \dots = L_{b_V} = 0$ have coefficient matrix

$$\begin{pmatrix} 1 & b_1 & \dots & b_1^{V-1} \\ \vdots & \vdots & & \vdots \\ 1 & b_V & \dots & b_V^{V-1} \end{pmatrix},$$

which is Vandermonde and invertible. Hence the choice gives one solution.

At that solution, the monic polynomial

$$T^V + X_V T^{V-1} + \dots + X_2 T + X_1$$

has $b_1, \dots, b_V$ as its roots. It therefore vanishes at none of the V unchosen nodes. Consequently the solution satisfies exactly one factor in each product g_i , and different choices give different solutions. Conversely, any common zero of the products must choose at least one factor from each pair. It cannot satisfy both factors in one pair, because that would give at least $V + 1$ distinct roots of the displayed monic polynomial. Thus there are exactly 2^V roots.

At a root, row i of the Jacobian of g is the Vandermonde row belonging to the chosen node, multiplied by the nonzero value of the unchosen factor. The Jacobian is therefore an invertible Vandermonde matrix times an invertible diagonal matrix. Enumerating the 2^V choices and forming the corresponding degree- V polynomials gives the stated $O(V^2C)$ bound. $\square$

This construction uses no division by an integer and no condition on the characteristic beyond the existence of $2V$ distinct field elements. For the QR-UOV working extensions below, this condition is trivial.

7.3 A full random separator

A primitive linear form must separate finitely many branch values and avoid finitely many leading-coefficient vectors. The published proof draws from a short integer-indexed family; that is the source of its second characteristic bound. Sampling the entire dual space is simpler over a finite extension.

Lemma 7.2 (Separator over a finite extension). Let A be a domain containing the finite field k' , and let $z_1, \dots, z_s \in A^V$ be nonzero. For a uniform $\lambda = (\lambda_1, \dots, \lambda_V) \leftarrow (k')^V$,

$$\Pr[\lambda \cdot z_j = 0 \text{ for some } j] \leq \frac{s}{|k'|}. \quad (41)$$

In particular, if $s \leq C^2$, the separator succeeds with probability at least $1 - C^2/|k'|$.

Proof. Fix a nonzero vector z . Choose an index h with $z_h \neq 0$. After the other $V - 1$ coefficients of λ are fixed, at most one value of $\lambda_h \in k'$ can make $\lambda \cdot z = 0$: two such values would give a nonzero scalar in k' annihilating the nonzero domain element z_h . Thus one vector is killed with probability at most $1/|k'|$. A union bound proves the claim. $\square$

Safey El Din and Schost identify at most C^2 vectors that a well-separating form must avoid: differences between start roots, differences between relevant target specializations, and leading-coefficient vectors of branches [34, Lem. 14 and Rem. 15]. [Theorem 7.2](#) therefore replaces their integer family without changing the success criterion.

7.4 The lifting identities do not require large characteristic

For completeness, we isolate the two local identities used by precision doubling. They work over a commutative ring and do not divide by factorials or by the characteristic.

Lemma 7.3 (Newton–Hensel precision doubling). Let R be a commutative ring, let $I \subset R$ be an ideal, and let $F \in R[X_1, \dots, X_V]^V$. Suppose

$$F(x) \equiv 0 \pmod{I^m}$$

and that $J_F(x)$ is invertible modulo I^m . Put

$$x^+ := x - J_F(x)^{-1}F(x) \pmod{I^{2m}}. \quad (42)$$

Then $F(x^+) \equiv 0 \pmod{I^{2m}}$.

Proof. The correction $h = x^+ - x$ lies in I^m . Polynomial Taylor expansion gives

$$F(x+h) = F(x) + J_F(x)h + R_2(x, h),$$

where every monomial in R_2 contains at least two coordinates of h . Hence $R_2(x, h) \in I^{2m}$. The first two terms cancel by (42). No denominator is used. $\square$

Lemma 7.4 (First-order change of primitive parameter). Work modulo T^{2m} . Let $Q(T, U)$ be monic in U , let $X(T, U) \in (R[T, U]/(T^{2m}, Q))^V$, and let λ be linear. Suppose

$$e(T, U) := \lambda(X(T, U)) - U \in T^m R[T, U]/(T^{2m}, Q).$$

Define

$$X^* := X - (\partial_U X)e, \quad Q^* := Q - (\partial_U Q)e. \quad (43)$$

Then the change of parameter $U_{\text{old}} = U - e(T, U)$ represents the same lifted branches modulo T^{2m} , and

$$\lambda(X^*) \equiv U \pmod{(T^{2m}, Q^*)}. \quad (44)$$

Proof. Because $e \in T^m$, substitution of $U - e$ into any polynomial is, modulo T^{2m} , its first-order Taylor expansion. This gives exactly (43). Moreover

$$\lambda(X^*) - U = e - (\partial_U \lambda(X))e = -(\partial_U e)e.$$

Differentiation in U does not change T -adic order, so $\partial_U e \in T^m$ and the product lies in T^{2m} . Again, the argument uses no integer denominator. $\square$

7.5 Proof crosswalk for the finite-characteristic solver

Proposition 7.5 (Where the published characteristic bound is used). For a square system of quadrics over a finite perfect field, replace the start construction in Safey El Din–Schost Lemma 7 by [Theorem 7.1](#), and replace the separator family in their Lemma 14 by [Theorem 7.2](#). The remaining correctness proof of their NONSINGULARSOLUTIONS algorithm applies over the extension field without a further lower bound on the characteristic.

Proof. We follow the proof in the order in which its ingredients are used.

Start fiber. The original lower bound $\text{char } k \geq \sum_i d_i$ ensures that its integer-indexed affine forms are distinct and that the product start has the full multihomogeneous root count with invertible Jacobian. [Theorem 7.1](#) supplies the same three facts directly: degree two, exactly C explicit roots, and a simple start fiber.

Homotopy curve. The homotopy is

$$H(T, X) = (1 - T)g(X) + Tf(X).$$

The published construction saturates by the X -Jacobian determinant. Lazard’s lemma and the multihomogeneous degree bound then give a one-dimensional curve of degree at most C_0 . These arguments require the simple start just established, not an ordering of integers inside the field.

Target specialization. The target system can have fewer than C isolated nonsingular roots and may have singular or positive-dimensional components. The published proof handles this by generalized power series, explicitly because ordinary Puiseux series are not sufficient in arbitrary characteristic [\[34, Sec. 3.4\]](#). Its valuation and specialization lemmas therefore already cover the finite-field setting.

Precision doubling and primitive normalization. The global lifting step is Newton–Hensel lifting of a geometric resolution. Its local identities are [Theorems 7.3](#) and [7.4](#); interpolation and polynomial arithmetic use inverses guaranteed by the simple fiber and separating form. No factorial or characteristic-dependent scalar is introduced.

Separator and cleaning. The published well-separating condition is avoidance of at most C^2 nonzero vectors. [Theorem 7.2](#) supplies it with probability at least $1 - C^2/|k'|$. Finally, the published CLEAN call removes every point where the target Jacobian vanishes. As noted in its Remark 15, every output is therefore a subset of the nonsingular target roots; a bad separator may omit roots or cause failure, but cannot introduce an invalid point.

These replacements cover the two uses of the integer-indexed finite families. All other steps are field algebra, generalized-power-series specialization, or Jacobian cleaning over the perfect extension field. $\square$

Theorem 7.6 (Nonsingular roots in characteristic 127). Let k be a finite field and let $f = (f_1, \dots, f_V)$ be affine quadrics over k , represented by a straight-line program of size L . Let k'/k be a finite extension containing at least $2V$ elements. There is a randomized algorithm with the following properties.

1. Every returned point belongs to $Z_{\text{ns}}(f)$.
2. Every point of $Z_{\text{ns}}(f)$ is represented in the output with probability at least

$$1 - \frac{C^2}{|k'|}. \tag{45}$$

3. The arithmetic cost is

$$\tilde{O}\left(CC_0(L + 2V + V^2)V\right) \tag{46}$$

operations in k' .

In particular, $|k'| \geq 8C^2$ gives one-invocation inclusion probability at least $7/8$.

Proof. Run the symbolic-homotopy algorithm of Safey El Din and Schost with the start from [Theorem 7.1](#) and a uniform full separator from [Theorem 7.2](#). [Theorem 7.5](#) gives correctness over k' . The well-separating event has probability at least [\(45\)](#); on that event the published specialization

recovers every isolated nonsingular target branch. The final CLEAN step proves item 1 on every branch, including branches with a bad separator.

For V quadrics in one block, the multihomogeneous quantities in the published Proposition 5 are exactly C and C_0 from (38); the sum of degrees is $2V$. The new start costs $O(V^2C)$ and does not change the dominant lifting term. Substitution into that proposition gives (46). $\square$

Remark 7.7 (One-sided failure is enough for cryptanalysis). The original solver paper notes that it cannot cheaply certify completeness of a generic output. QR-UOV does not need such a certificate. We impose every unrecombined directional equation, require a K -rational root, complete the graph, and verify the unchanged public map. An omitted planted root causes a restart. An extraneous or malformed candidate is rejected.

7.6 End-to-end verified directional recovery

For a random $V \times m$ matrix R over K , put

$$f_{c,R} := R(q_{1,c}, \dots, q_{m,c})^\top.$$

If $\mathcal{B}(c)$ has rank V , then $R\mathcal{B}(c)$ is a uniform $V \times V$ matrix. Write

$$\mu_V(Q) := \prod_{j=1}^V (1 - Q^{-j}), \quad Q = |K| = 127^3. \quad (47)$$

Choose $K' = \mathbb{F}_{Q^r}$ with

$$r := \min\{s \geq 1 : Q^s \geq 8 \cdot 2^{2V}\}. \quad (48)$$

The values are

$$r = 6, \quad 8, \quad 10 \quad (49)$$

at Levels I, III, and V.

Theorem 7.8 (Verified directional equivalent-key recovery). Let a fixed expanded QR-UOV public key have a valid graph decomposition G . Suppose

1. some maximal minor of $\mathcal{B}(c)$ is a nonzero polynomial in c ; and
2. the recovered planted decomposition has a nonzero signing determinant polynomial Δ from Theorem 3.7.

Then the following Las Vegas algorithm returns a publicly verified functional equivalent key.

1. Sample $c \leftarrow K^O$ and $R \leftarrow K^{V \times m}$.
2. Run Theorem 7.6 on $f_{c,R}$ over K' from (48).
3. Intersect the returned parametrization with all m original directional equations and with the K -rational locus.
4. For every retained root x , run the graph completion of Theorem 5.2; require all graph identities, the inverse pull-back, exact public-map factorization, and a signing witness.
5. Return the first accepted key; otherwise restart with fresh public coins.

One invocation includes the planted branch with probability at least

$$p_{\text{dir}} \geq \left(1 - \frac{V}{Q}\right) \mu_V(Q) \left(1 - \frac{2^{2V}}{Q^r}\right). \quad (50)$$

The expected numbers of full solver invocations are at most

$$1.0000262, \quad 1.0000561, \quad 1.0202201. \quad (51)$$

Every returned key is correct, regardless of whether failed solver invocations were complete.

Proof. By [Theorem 5.4](#), a random direction has full planted derivative with probability at least $1 - V/Q$. Conditional on that event, $R\mathcal{B}(c)$ is uniform square, so it is invertible with probability $\mu_V(Q)$. The planted point is then an isolated nonsingular root of $f_{c,R}$ by [Theorem 5.1](#). [Theorem 7.6](#) includes it with probability at least $1 - 2^{2V}/Q^r$.

The all-equation and rational-root filters retain the planted point. [Theorem 5.2](#) reconstructs G , and [Theorems 3.4](#) and [3.7](#) supply the exact factorization and signing certificate. Conversely, any candidate that fails an original equation, completion equation, graph identity, pull-back identity, factorization check, or oil-matrix test is rejected. This proves zero-error acceptance. Fresh restarts and (50) give the geometric tail and the numerical expectations. $\square$

Corollary 7.9 (Public-random key density). In the public-random expanded-key model, the probability that every coordinate direction has deficient planted derivative is $\tau_{m,V}(Q)^O$, with base-two logarithms

$$-1132.166907, \quad -1635.352198, \quad -2935.247544.$$

The probability that the deterministic pair-line signing search fails has base-two logarithm below

$$-544.820, \quad -796.276, \quad -1068.687.$$

Outside their union, a deterministic scan of coordinate directions and signing witnesses contains a successful outer branch; only the internal nonsingular-root solver requires random coins.

Proof. Combine [Theorems 3.7](#) and [5.4](#). The two exceptional events need not be independent; a union bound suffices. $\square$

8 Concrete solver refinements and the Level-I estimate

The generic root solver does not by itself fix the concrete cost. This section first states the resulting candidate bound. It then gives the arithmetic choices and branch filters used by the final ledger: batched dense lifting, sparse working fields, local separation, and exact Frobenius descent. The section ends with the root-success bound and the finite Level-I ledger.

8.1 Candidate local-separator bound

Let $q = 127$, $a = m - 4$, $C = 2^a$, and $C_0 = 2^{a-1}(a + 2)$ as in the parent attack. The supplied audit optimizes the extension degree over certified sparse irreducible polynomials.

Theorem 8.1 (Candidate local-separator bound). Assume the public-random model, homotopy degree bound, and Boolean-gate conventions of the preceding direct-attack construction and the split-before-lift refinement developed here. Replace global separation by the local conditions in [Section 8.5](#), accept only simple linear target factors, and use the exact descent test in [Section 8.6](#). Then the retry-aware diagnostic costs are those in [Table 1](#). In particular, Level I costs

$$2^{140.383995063}$$

Boolean gates, leaving 2.616004937 bits below the category target.

Table 1: Audited stress estimates. “Global split” is the preceding split-before-lift bundle.

Level	Parent	Global split	Local split	Target	Margin
I	151.639	141.126	140.384	143	+2.616
III	203.507	191.630	191.027	207	+15.973
V	260.389	247.637	247.125	272	+24.875

The Level-I optimum in the main ledger uses $K' = \mathbb{F}_{127^8}$ with the Rabin-certified modulus

$$X^8 + X^4 - 1.$$

Degree 9, using $X^9 + X^2 + 1$, gives total 140.418965 and is retained as a nearby cross-check.

8.2 Dense-quadratic batching and flattened lifting

8.2.1 Why the direct straight-line-program bound is not the last word

For dense quadrics, a direct straight-line program has size $L = O(V^3)$. Substituting it into (46) gives

$$\tilde{O}(V^5 4^V) \quad (52)$$

operations in K' . This is a valid generic specialization, but it repeatedly charges contractions that all use the same dense quadratic coefficient tensor.

Write

$$f_i(X) = X^\top A_i X + b_i^\top X + c_i. \quad (53)$$

At precision k , the lifted coordinates live in a coefficient algebra

$$\mathcal{A}_k := K'[T, U]/(T^k, Q_k(T, U)), \quad (54)$$

where Q_k is monic of U -degree at most C . The V coordinate vectors form a $V \times (Ck)$ coefficient matrix. Flattening the matrices A_i turns all Jacobian contractions into one rectangular matrix product over K' . Quadratic values follow from the Jacobian and X , while the Newton solve uses matrix algebra over $\mathcal{A}_k$.

Proposition 8.2 (Dense-quadratic lifting ledger). Let ω be an admissible square-matrix multiplication exponent, and let $\mu(C, k)$ be the cost of one multiplication in $\mathcal{A}_k$. The precision- k work of the global Newton lift can be organized using

$$\tilde{O}(V^3 Ck + (V^\omega + V^2)\mu(C, k)) \quad (55)$$

operations in K' . Summed over precision doubling through $2C_0$, this gives

$$\tilde{O}(V^3 C C_0 + (V^\omega + V^2)\mu(C, 2C_0)). \quad (56)$$

Proof. Stack the V matrices A_i into a $V^2 \times V$ scalar matrix. Multiplication by the $V \times (Ck)$ lifted-coordinate matrix forms every linear Jacobian contraction simultaneously in $O(V^3 Ck)$ scalar operations. This term is written explicitly rather than hidden inside the coefficient-algebra multiplication count.

The identity

$$f_i(X) = \frac{1}{2}X^\top (2A_i X + b_i) + \frac{1}{2}b_i^\top X + c_i$$

computes all values with $O(V^2)$ products in $\mathcal{A}_k$. Newton matrix inversion or solving costs $O(V^\omega)$ such products, and primitive-element renormalization lies within the same envelope. The terminal precision is $2C_0$; quasi-linear multiplication makes the last doubling stage dominate up to logarithmic factors. $\square$

8.2.2 One multiplication, not a product of two multiplication costs

The earlier stress analysis bounded multiplication in (54) by applying a quasi-linear polynomial-multiplication bound first in U and then in T . This gives a product $M(C)M(k)$. A bivariate array can instead be flattened once.

Lemma 8.3 (Collision-free Kronecker flattening). Let $R_k = K'[T]/(T^k)$ and let $Q_k(U) \in R_k[U]$ be monic of degree C . After precomputing the reciprocal of the reversed Q_k , one multiplication in $R_k[U]/(Q_k)$ can be performed using at most three ordinary polynomial multiplications whose input length is below $2Ck$, plus $O(Ck)$ additions. The reciprocal precomputation costs $O(M(2Ck) \log C)$ operations in K' .

Proof. For a polynomial

$$A(U, T) = \sum_{0 \leq i < C, 0 \leq j < k} a_{i,j} U^i T^j,$$

choose the radix $B = 2C - 1$ and encode

$$U^i T^j \mapsto Z^{i+Bj}. \quad (57)$$

In a product, the U -exponent is at most $2C - 2 = B - 1$. It therefore never carries into the T digit. The largest input exponent is $(C - 1) + B(k - 1) < Bk < 2Ck$, so one ordinary univariate multiplication recovers the complete bivariate product modulo T^k .

To reduce the U -degree modulo the monic polynomial Q_k , use the standard reciprocal division algorithm over the coefficient ring R_k . With the reciprocal precomputed, one product obtains the quotient estimate and one product forms the quotient times Q_k . Together with the unreduced product, this is three products of the same padded length. Newton inversion of the reversed monic polynomial gives the stated one-time reciprocal cost over any commutative coefficient ring. $\square$

Corollary 8.4 (Flattened lifting complexity). Let $M(n)$ denote a quasi-linear polynomial-multiplication cost. The batched lift can be implemented in

$$\tilde{O}(V^3 C C_0 + (V^\omega + V^2) M(C C_0)) = \tilde{O}(V^{\omega+1} 4^V) \quad (58)$$

operations in K' .

For the explicit stress function

$$M(n) := n \log_2 n \log_2 \log_2 n, \quad (59)$$

the entire precision-doubling schedule for one coefficient-algebra product is bounded by

$$3M(8CC_0). \quad (60)$$

Proof. Apply Theorem 8.3 at precisions $2C_0, C_0, C_0/2, \dots$. At the terminal stage, each encoded input has length below $4CC_0$. Monotonicity of the logarithmic factors gives

$$\sum_{j \geq 0} M\left(\frac{4CC_0}{2^j}\right) < 2M(4CC_0) < M(8CC_0).$$

There are three ordinary products per modular multiplication. Reciprocal precomputation contributes an additional logarithmic factor but is dominated by the explicit V^2 outer allowance. Finally, $CC_0 = 2^{2V-1}(V + 2)$. $\square$

Remark 8.5 (What was improved and what was corrected). Flattening does not alter the exponential factor 4^V ; it replaces the product of two independent quasi-linear overheads by the overhead at the total coefficient-array length. The factor 8 in (60) is important: an earlier draft stopped at precision C_0 , whereas rational reconstruction requires $2C_0$. All finite values in Section 10 use the corrected precision.

8.3 Sparse working-field models

Both solvers may choose any convenient representation of their finite working field. The directional solver starts over $K = \mathbb{F}_{127^3}$, so its auxiliary fields must contain K ; the direct solver starts over $\mathbb{F}_{127}$ and has no such divisibility requirement. Projectivization lowers the Level-I direct working degree from 16 to 15.

Lemma 8.6 (Explicit sparse auxiliary fields). The following polynomials are irreducible over $\mathbb{F}_{127}$:

$$\begin{aligned}\phi_{15}(X) &= X^{15} + X^{12} - X + 1, & \phi_{18}(X) &= X^{18} + X^4 + 1, \\ \phi_{22}(X) &= X^{22} + X + 1, & \phi_{24}(X) &= X^{24} + X^5 - X + 1, \\ \phi_{30}(X) &= X^{30} + X^7 + 1.\end{aligned}$$

The projective direct-forgery solver uses degrees 15, 22, 30 at Levels I, III, and V. The directional solver uses degrees 18, 24, 30.

Proof. The companion audit applies Rabin's irreducibility criterion [33]. For every displayed degree d , it verifies

$$X^{127^d} = X \pmod{\phi_d}$$

and, for each prime divisor e of d ,

$$\gcd(\phi_d, X^{127^{d/e}} - X) = 1.$$

All computations are exact over $\mathbb{F}_{127}$, and the machine-readable output records every gcd degree. The same audit certifies the additional sparse fields used in the retuning table of Table 7, including $X^{26} + X^3 - X + 1$. $\square$

Schoolbook convolution is unnecessary at these modest extension degrees. The next statement freezes an explicit recursive circuit rather than invoking an asymptotic multiplication theorem.

Proposition 8.7 (Audited Karatsuba extension-multiplication envelope). Let

$$g_{\text{mul}} = G_{\text{mul}}(\log_2 127), \quad g_{\text{add}} = 4 \log_2 127.$$

For length- d coefficient vectors, let (M_d, A_d) be chosen recursively as follows. The schoolbook candidate is

$$(d^2, (d-1)^2).$$

For $d \geq 2$, put $a = \lceil d/2 \rceil$ and $b = \lfloor d/2 \rfloor$. The Karatsuba candidate is

$$(2M_a + M_b, 2A_a + A_b + 4d - 4). \tag{61}$$

Choose whichever candidate has smaller value $Mg_{\text{mul}} + Ag_{\text{add}}$. If s_d is the number of nonleading terms of ϕ_d , one multiplication in $\mathbb{F}_{127^d}$ followed by sparse reduction is then charged by

$$G_{127^d}^{\text{sparse}} := M_d g_{\text{mul}} + (A_d + s_d(d-1))g_{\text{add}}. \tag{62}$$

For the official working fields, the selected plans are:

d	M_d	A_d	s_d
15	81	332	3
18	153	404	2
22	225	480	2
24	243	512	3
30	243	1112	2

Proof. The schoolbook counts are exact: d^2 coefficient products and $d^2 - (2d - 1) = (d - 1)^2$ additions. For the recursive candidate, write each operand as a low block of length a plus X^a times a high block of length b . The products of the two low blocks and of their sums have length a ; the high-block product has length b , giving $2M_a + M_b$ coefficient multiplications. Forming the sums, subtracting the low and high products from the middle product, and combining the three shifted outputs uses at most $4d - 4$ additional coefficient additions, giving (61). Recursion therefore realizes the displayed pair (M_d, A_d) .

The unreduced product has only $d - 1$ high coefficients. Eliminating each one modulo ϕ_d uses s_d additions or subtractions and no nontrivial constant multiplication because every nonleading coefficient is $+1$ or -1 . This proves (62). The companion audit executes the same recursive plan and compares 48 random products against schoolbook convolution before certifying the gate counts. $\square$

8.4 Characteristic polynomials without global separation

Let $\Gamma \subset \mathbb{A}^{1+a}$ be the union of the saturated homotopy-curve components that dominate the t -line, with degree at most $D = C_0$. The Jacobian saturation removes the non-etale locus, so over $K'(t)$ its generic fiber is a finite product of separable field extensions,

$$B = K'(t)[X_1, \dots, X_a]/J, \quad \dim_{K'(t)} B = C.$$

Write x_j for the image of X_j and M_{x_j} for multiplication by x_j on B .

Introduce independent coefficients $\Lambda = (\Lambda_1, \dots, \Lambda_a)$ and the generic linear element

$$u_\Lambda = \sum_j \Lambda_j x_j.$$

Its characteristic polynomial is

$$\chi_\Lambda(t, U) = \det \left(UI - \sum_j \Lambda_j M_{x_j} \right). \quad (63)$$

Schost proves two facts that are exactly suited to the present use: specialization of Λ gives the characteristic polynomial of the specialized linear element, and the generic characteristic polynomial can be written

$$\chi_\Lambda = \frac{\Xi_\Lambda}{\Theta},$$

where the t -degrees of Ξ_Λ and Θ are at most $\deg \Gamma \leq D$ and Θ is independent of Λ [35, Lemmas 2–3 and Proposition 2]. This is the characteristic-polynomial, or generic u -resultant, form of the parametric degree theorem.

Schost phrases one corollary using a primitive element with coefficients in the ground field, together with a field-size hypothesis. The generic characteristic polynomial itself does not require such an element. After extending to an algebraic closure of $K'(t)$, the finite etale algebra has C distinct geometric points. For every distinct pair, the difference

$$\Lambda \cdot (x_s - x_r)$$

is a nonzero polynomial in the independent indeterminates Λ . Hence u_Λ separates the geometric points over the rational-function field $K'(t)(\Lambda)$ and is primitive there, irrespective of the size of K' . Apply Schost's degree argument after this transcendental scalar extension. The determinant in (63) is formed from the original multiplication matrices, so its coefficients already lie in $K'(t)[\Lambda, U]$; scalar extension cannot lower a reduced t -degree and thereby hide a larger degree over K' . Thus no K' -rational global separator, and in particular no $|K'| \gg C^2$ assumption, is used.

Lemma 8.8 (Nonseparating eliminant). For every $\lambda \in K'^a$, including a colliding one,

$$q_\lambda(t, U) := \prod_{s=1}^C (U - \lambda(x_s(t))) = \chi_\Lambda(t, U)|_{\Lambda=\lambda}.$$

Every coefficient of q_λ has a common-denominator representation with numerator and denominator t -degrees at most D .

Proof. The first identity is the characteristic-polynomial product formula in the reduced generic fiber, with repeated factors allowed after specialization. The second assertion follows by substituting $\Lambda = \lambda$ in Ξ_Λ/Θ . Since Θ is independent of Λ , a separator collision cannot create a new parameter denominator or increase its degree. $\square$

The coordinate interpolation numerators are already encoded in the same generic characteristic polynomial.

Lemma 8.9 (Characteristic-polynomial polar). Define

$$v_j(t, U) = \sum_s x_{s,j}(t) \prod_{r \neq s} (U - \lambda(x_r(t))).$$

Then

$$v_j(t, U) = - \left. \frac{\partial \chi_\Lambda(t, U)}{\partial \Lambda_j} \right|_{\Lambda=\lambda}. \quad (64)$$

Consequently each v_j , and every fixed affine-linear combination of the v_j and q_U , has numerator and denominator t -degrees at most D , without assuming that λ separates the branches.

Proof. Over a splitting field, differentiate

$$\chi_\Lambda(t, U) = \prod_s \left(U - \sum_i \Lambda_i x_{s,i}(t) \right)$$

with respect to Λ_j and then specialize $\Lambda = \lambda$. This gives (64). Differentiating Ξ_Λ in the linear-form coefficients does not increase its t -degree, while the denominator Θ is unchanged. $\square$

This directly covers the sole linear sketch used later:

$$b_1(P) = \lambda^{q^{d-1}} \cdot P.$$

Its interpolation numerator is the corresponding fixed K' -linear combination of the v_j .

For the rational oil filter

$$\psi(X) = \frac{H_0(X)}{c + \lambda(X)},$$

the preceding split-before-lift note proves that the saturated graph curve has degree at most $2D$. Apply the same construction to its generic linear element

$$u_{\Lambda, \mu} = \Lambda \cdot X + \mu Y,$$

where $Y = \psi(X)$ is the graph coordinate. Specializing $(\Lambda, \mu) = (\lambda, 0)$ gives q_λ , and

$$- \left. \frac{\partial \chi_{\Lambda, \mu}}{\partial \mu} \right|_{(\lambda, 0)} = v_\psi.$$

The graph degree bound therefore gives numerator and denominator t -degrees at most $2D$ for the filter numerator. Consequently the existing precision

$$P = 4C_0 + 1$$

remains valid.

Remark 8.10. Separation is needed only when one wants the linear element to be a primitive coordinate for every point. The attack instead keeps the full characteristic polynomial and later accepts only a simple target factor. Characteristic polynomials and their coefficient polars remain defined at colliding linear forms.

8.5 Local separation at the target point

At target specialization, normalize q and each scalar numerator by the common t -valuation exactly as in Safey El Din–Schost Lemma 12 [34]. Call λ *valuation-generic* if every branch escaping to infinity with minimum coordinate valuation $\mu_s < 0$ satisfies

$$\text{ord } \lambda(x_s) = \mu_s.$$

This condition, not pairwise branch separation, is what the specialization proof uses to remove the infinite factors from the specialized eliminant.

Lemma 8.11 (Simple-factor recovery). Assume λ is valuation-generic. Let $\bar{q}(U)$ and $\bar{v}_b(U)$ be the normalized target specializations. If $\tau \in K'$ is a simple root of $\bar{q}$, then there is exactly one finite represented target point P with $\lambda(P) = \tau$, and

$$b(P) = \frac{\bar{v}_b(\tau)}{\bar{q}'(\tau)}.$$

No separation condition is needed away from that one fiber value.

Proof. Let S_∞ and S_0 index the escaping and finite branches. With

$$e = - \sum_{s \in S_\infty} \mu_s,$$

valuation genericity gives

$$(t^e q)(0, U) = \gamma \prod_{s \in S_0} (U - \lambda(P_s))$$

for a nonzero scalar γ . In the normalized numerator, the summand indexed by a finite branch s specializes to

$$\gamma b(P_s) \prod_{r \in S_0, r \neq s} (U - \lambda(P_r)).$$

A summand indexed by an escaping branch need not vanish term by term; rather, every such surviving summand is divisible by the full finite product $\prod_{r \in S_0} (U - \lambda(P_r))$. It therefore vanishes modulo $\bar{q}$, and in particular at every finite root of $\bar{q}$. Thus $\bar{v}_b$ modulo $\bar{q}$ is the ordinary interpolation numerator on the finite limits. A simple root τ belongs to exactly one finite branch, and evaluation followed by division by $\bar{q}'(\tau)$ gives the claimed value. $\square$

Condition on the existence of a canonical eligible nonsingular base-field target point $P_\star$; for example, choose the first under any fixed ordering. Nonsingularity makes it an isolated multiplicity-one point. Let I generic branches escape to infinity and let F have finite target limits, counted with branch multiplicity. Then $I + F = C$. There are at most I valuation-cancellation hyperplanes and at most $F - 1$ collision hyperplanes against $P_\star$. Each bad equality

$$\lambda(w) = 0 \quad \text{or} \quad \lambda(P - P_\star) = 0$$

cuts out at most $|K'|^{a-1}$ choices of λ in $(K')^a$. Thus a uniform $\lambda \in (K')^a$ fails valuation genericity or collides at $P_\star$ with probability at most

$$\frac{I + F - 1}{|K'|} = \frac{C - 1}{|K'|}. \tag{65}$$

The coefficient vector of the equation may lie only over an algebraic closure, so its solution set need not literally be a K' -defined hyperplane. The count is still valid: choose a nonzero coordinate of that vector and fix all other coefficients of λ ; the remaining equation has at most one solution in K' (and may have none).

For the filter denominator, first compute the C known start values $\lambda(\xi_s)$ and sample c uniformly from the complement of the forbidden set $\{-\lambda(\xi_s)\}_s$ in K' . Rejection sampling costs an expected factor at most $(1 - C/|K'|)^{-1}$ for a scan of the start values; this scan is negligible beside one length- P filter lift and is included in the miscellaneous envelope. Conditional on this choice, only the at most $F \leq C$ finite target values remain forbidden. Therefore

$$\Pr[c + \lambda(P) = 0 \text{ for a finite target branch}] \leq \frac{C}{|K'| - C}. \quad (66)$$

The old $C^2/|K'|$ event has disappeared.

8.6 Exact one-sketch Frobenius descent

Set

$$b_1(P) = \lambda^{q^{d-1}} \cdot P, \quad K' = \mathbb{F}_{q^d}.$$

The separator value $\tau = \lambda(P)$ is already available from the factor of $\bar{q}$, so no numerator for $\lambda \cdot P$ is needed.

Lemma 8.12 (Simple-root descent is exact). Let $\tau \in K'$ be a simple linear factor of the target eliminant, and let P be its unique represented point. Then

$$b_1(P)^q = \tau \iff P \in \mathbb{F}_q^a.$$

Proof. The forward identity for a base-field point is immediate. Conversely,

$$b_1(P)^q = \lambda \cdot P^q.$$

If it equals $\tau = \lambda \cdot P$, then P and P^q have the same separator value. The target equations and homotopy are defined over $\mathbb{F}_q$, so the represented finite target limits are stable under Frobenius. If $P^q \neq P$, the two distinct represented points produce multiplicity at least two at $U = \tau$, contradicting simplicity. Hence $P^q = P$. $\square$

This eliminates the probabilistic $C/|K'|$ false-positive term. It also eliminates the abort cap: every simple factor passing the Frobenius test and the nonzero-square oil filter is a genuine base-field target solution. The algorithm reconstructs coordinates for the first such factor and stops, so exactly one coordinate re-lift is charged per successful forgery.

8.7 Root success probability

For completeness, this pass expands the probability calculation inherited from the split-before-lift note. Let M count all off-oil affine roots of the normalized residual system, and let N count those with invertible affine Jacobian. The parent manuscript proves, for an actual parameter $\mu \geq 1 - q^{-1}$,

$$\mathbb{E}[M] = \mu, \quad \mathbb{E}[M^2] = 2\mu + \mu^2 - (q-1)\mu q^{-r}, \quad \mathbb{E}[N] = \mu\pi,$$

where $\pi = \prod_{j=1}^{r-1} (1 - q^{-j})$. The roots occur in pairs $z, -z$, and each nonsingular normalized pair yields one eligible projective point. Applying the second Bonferroni inequality to the projective-point events and bounding pairs of nonsingular roots by pairs of all roots gives

$$\begin{aligned} \Pr[\text{eligible nonsingular projective root}] &\geq \frac{\mathbb{E}[N]}{2} - \mathbb{E}\left[\binom{M/2}{2}\right] \\ &= \frac{\mu\pi}{2} - \frac{\mu^2}{8} + \frac{(q-1)\mu q^{-r}}{8}. \end{aligned}$$

The right-hand side is increasing for $\mu \in [1 - q^{-1}, 1]$, so substituting $\mu = 1 - q^{-1}$ yields

$$p_{\text{root}} = 0.3690869822 \dots$$

As a separate fallback, the ledger also substitutes the parent manuscript's already-proved lower bound $0.326336998767509 \dots$; the Level-I total then becomes 140.531461 , still 2.469 bits below target.

8.8 Finite Level-I ledger

At Level I the local variant has two scalar families, b_1 and ψ , instead of three. The terminal product tree therefore uses

$$\frac{a}{2}(1 + 2 \cdot 2) = 125$$

long-product units. Rational reconstruction contributes 99.885 units. Candidate coordinate evaluation is charged once over the complete Las Vegas execution, rather than per trial, and the 64 -unit miscellaneous cushion is retained. This pass also charges the one-time formation of $u_s = \lambda \cdot x_s$ during that coordinate rerun.

Table 2: Level-I component ledger for extension degree $d = 8$.

Component	$\log_2$ gates
Initial base-field branch lifts	139.358
Valuation-safe filter series	132.120
Terminal scalar recombination	138.791
One on-demand coordinate re-lift	137.860
Extension-valued leaf formation	127.610
Coordinate-rerun separator leaves	124.898
Factorization and exact verification envelope	115.707
Preprocessing	63.365
Total	140.383995

The one-trial success lower bound is 0.353997 , giving restart factor 2.824886 . At $d = 8$, the local-separator failure bound in (65) is 0.016637 , and the conditional target-denominator bound in (66) is 0.016918 .

Table 3: Level-I sensitivity to the multiplier on $M(n) \log n$ in rational reconstruction, retaining $d = 8$.

Multiplier	8	10	16	32	64	128
Total	140.384	140.424	140.538	140.805	141.225	141.815

Table 4: Level-I cross-checks. Each row reoptimizes the extension degree.

Online-envelope multiplier	1	2	4	8	16
Best extension degree	8	8	9	9	9
Total	140.384	141.117	141.952	142.856	143.806

Thus the candidate remains below 2^{143} throughout the rational-reconstruction sensitivity range. It also remains below target after multiplying the entire claimed online-product envelope by eight, but not by sixteen; the finite margin therefore does not excuse the remaining obligation to provide a fully explicit schedule. The aggregate Level-I margin corresponds to an overall multiplicative cushion of $2^{2.616} = 6.13$ in the inherited gate model; this does not address memory traffic or a change of model.

9 Evidence and limitations

The direct and directional reductions, the fixed-output scissors theorem, the determinant identities used by the local separators, and the exact Frobenius filter are algebraic results. They do not depend on the final Boolean-gate convention. Every reconstructed candidate is checked against the original public equations or the unchanged public verifier. A failed solver invocation may therefore force a retry, but it cannot produce an accepted false signature or equivalent key.

The reduced implementation shows that the local-separator attack composes at toy scale. It preserves the QR-UOV public quadratic relation but simplifies seed expansion, byte serialization, and the exact domain-separated hash interface. Starting from expanded public matrices, a message, and a salt, it runs the complete attack and produces a signature accepted by the public verifier. A public-only replay reproduces the result. The canonical run and five separate regressions give six successful $q = 13$ forgeries and replays. One supplementary $q = 29$ instance also forged and replayed successfully. This closes the composability gap between the individual algebraic checks, but it does not establish the full-parameter Boolean-gate estimate. The prototype uses naive truncated-series arithmetic at $a = 2$, so it neither implements nor benchmarks the fast multiplication schedule used in that estimate. The full experiment record appears in [Section I](#).

That estimate remains conditional on four points. First, the characteristic-polynomial degree theorem must apply to the exact saturated dominant homotopy algebra and its rational-filter graph with the stated degree bounds. Second, the fast online-product envelope needs a line-by-line full-scale schedule. Third, the public-random model must transfer to the seeded public-key expansion. Finally, the gate ledger does not account for the astronomical memory traffic of the symbolic representation. No defect is currently known in the determinant identities, but the Level-I margin is only 2.616 bits, so these obligations are material. The technical supplement retains the complete validation record and executable-audit inventory.

Table 5: Status of the main claims.

Claim	Status
Projective direct core $m - 4$	Exact for every nonzero target once a seed root is supplied; Chevalley–Warning guarantees such a root in every seven-dimensional seed, and homogeneity quotients the residual by scaling.
Seed root is off oil	Fixed-key probability bound over public attack coins; no random-key assumption.
Common-isotropic extension	Deterministic and finite after the seed, using linear algebra and exhaustive search in $\mathbb{P}^6(\mathbb{F}_q)$.
Eligible projective-root probability	Exact lower bound in the independent-graph public-random degree-three model.
One regular direction determines an equivalent key	Exact for every expanded key satisfying the public rank and signing checks.
Returned signature or key is valid	Exact; every candidate is verified against the unchanged public map.
Finite cost rows	Reproducible symbolic-algorithm diagnostics, not measured circuits.
Full-parameter memory	Prohibitive for the direct representations analyzed here.
Fixed SHAKE/AES expansion	Key-by-key verification is exact; density statements transfer only in the stated ideal-primitive models, with explicit buffer and seed-collision terms.

10 Combined cost, memory, and parameter implications

10.1 Work unit and scope

The QR-UOV specification expresses classical attack costs as Boolean-gate exponents and charges multiplication of b -bit field elements by

$$G_{\text{mul}}(b) = 2b^2 + b \quad (67)$$

Boolean gates [3]. We retain that broad unit. Every extension-field operation is overcharged as one multiplication using the explicit Karatsuba circuit of (62).

The principal symbolic-homotopy term uses the corrected reconstruction precision $2C_0$. For a square solver core of dimension a ,

$$C = 2^a, \quad C_0 = 2^{a-1}(a + 2),$$

and the flattened lifting schedule is charged by

$$3(a^3 + a^2)M(8CC_0) \quad (68)$$

operations in the chosen working field. The projective direct row uses $a = m - 4$ and includes the fixed-key seed bound, nonsingular-root factor, random chart, and finite-field separator. The directional row uses $a = V$ and includes direction regularity and its separator. Polynomial-time preprocessing, all-equation filtering, factorization, graph completion, square testing and rescaling, and public verification are included through explicit lower-stage envelopes; they do not change the displayed decimals.

These are reproducible diagnostics for symbolic algorithms, not measured circuits. In particular, setting

$$M(t) = t \log_2 t \log_2 \log_2 t$$

assigns unit coefficient to an asymptotic multiplication bound. The exact reductions, Karatsuba circuits, sparse fields, and success probabilities do not depend on that convention; the final conversion to a practical implementation does.

Table 6: Classical gate diagnostics, including the local-separator split-before-lift refinement and the audited Karatsuba field circuit. A positive margin means that the attack row lies below the category target.

Attack	Level I		Level III		Level V	
	Cost	Margin	Cost	Margin	Cost	Margin
Original projective forgery	151.639	−8.639	203.507	3.493	260.389	11.611
Local-separator projective forgery	140.384	2.616	191.027	15.973	247.125	24.875
Directional equivalent-key recovery	154.826	−11.826	206.177	0.823	260.857	11.143
Best attack	140.384	2.616	191.027	15.973	247.125	24.875
Category target	143		207		272	

10.2 The two attacks

The original direct rows use projective solver dimensions 50, 74, 101. The local-separator refinement keeps those algebraic cores but lowers the working-field degrees and removes global branch separation, the second Frobenius numerator, probabilistic false candidates, and repeated coordinate recovery. Its optimized field degrees are 8, 12, and 16 in the supplied ledger. The directional rows remain an independent equivalent-key route. The reduced implementation executes the local-separator route at toy scale; the full-parameter symbolic lift still dominates both time and memory.

For context, the QR-UOV team’s January 2026 response reports classical Hashimoto estimates of 158, 217, 285 bits for the same three public dimensions [36]. The new direct rows are lower by approximately 6.36, 13.49, 24.61 bits. This is not a claim that the implementations are identical: the team uses a semi-regular direct-MQ model, whereas our row uses the finite-characteristic nonsingular-root solver and its explicit success probability.

10.3 What follows at each level

Level I. The local-separator row is 140.384, or 2.616 bits below the 143-bit target. Thus the paper gives a candidate Level-I break in the stated Boolean-gate model. This is not a practical attack claim: the symbolic state is physically unrealizable, and the full-scale online-product schedule and characteristic-polynomial degree transfer remain explicit obligations. The reduced-parameter implementation establishes that the algebraic pipeline is a single executable attack, not that the Level-I extrapolation is practical.

Level III. The local-separator row is 191.027, leaving 15.973 bits below target. The older projective and independent equivalent-key rows are also below target. The combined evidence strongly supports changing the Level-III parameters.

Level V. The local-separator row is 247.125, leaving 24.875 bits below target. The older projective and independent equivalent-key rows are likewise below target. Level V should not be retained unchanged.

10.4 Why changing only the vinegar count is not a repair

The official parameter sets sit on the direct side of the projective scissors crossing:

Table 7: Smallest quotient-compatible (V, m) pairs found by the companion audit that put both displayed attack rows at the category target, or at the target plus a 16-bit attack-specific cushion.

Level	Current		Reach target		Reach target +16	
	V	m	V	m	V	m
I	52	54	52	54	57	60
III	76	78	78	81	87	90
V	102	105	108	111	117	120

Level	V	$m - 4$	Directional core	Projective direct core
I	52	50	52	50
III	76	74	76	74
V	102	101	102	101

Increasing V raises the directional core but leaves the projective direct core unchanged. Decreasing V eventually makes directional recovery no larger than $m - 4$. Consequently a fixed- m vinegar retuning cannot move the combined algebraic bottleneck.

The next table gives attack-specific thresholds under the same padded ledger. It is not a complete replacement proposal: every other attack, failure probability, key size, and implementation cost must be recomputed. The point is that both columns must move. Raising only V leaves the direct attack unchanged; raising only m leaves the directional attack unchanged.

The table enforces $m \equiv 0 \pmod{3}$ and the direct-side feasibility condition $V \geq m - 3$. Level I already exceeds its bare target in this model, but a 16-bit cushion requires moving both dimensions. At Levels III and V, even the target-only diagnostic requires increases in both V and m ; a 16-bit cushion requires substantially larger changes. A structural redesign may be preferable to paying both costs.

10.5 Memory remains the central practical limitation

The desired outputs are small: one direct attack needs a single vector in $\mathbb{F}_q^n$, and the equivalent-key attack ultimately needs a $V \times O$ graph. The direct symbolic representations are not small. One flattened coefficient array at terminal precision requires approximately

$$2^{49.41} \text{ EiB}, \quad 2^{98.51} \text{ EiB}, \quad 2^{153.40} \text{ EiB}$$

for the projective direct cores. The corresponding directional arrays require about $2^{53.73}$, $2^{102.68}$, and $2^{155.41}$ EiB. A full-parameter execution using these representations is physically unrealistic.

This limitation prevents a claim of a practical full-parameter implementation. No full-parameter forgery or equivalent-key recovery is reported, and the paper draws no conclusion from unimplemented attack variants. The exact reductions and the fixed-output scissors theorem are algebraic results; time, storage, and correctness are reported as separate resources.

11 Conclusion

QR-UOV's projective direct-forgery and directional equivalent-key attacks obey a fixed-output algebraic ceiling. For every vinegar count and every nonzero verification target, one of the two attacks has nonlinear solver core dimension at most $m - 4$. Changing only the vinegar count therefore cannot eliminate both low-dimensional attack surfaces. A revision must also increase the output dimension or change the construction so that one of the reductions no longer applies.

The direct reduction gives square cores of dimensions 50, 74, 101, while the directional attack recovers an equivalent signing key. To estimate the cost of these cores, the local-separator refinement removes the need for global branch separation: characteristic-polynomial polars suffice for scalar recovery, and one exact Frobenius sketch identifies a base-field point on a simple target factor. Under the stated symbolic Boolean-gate model, the resulting rows lie below all three NIST category targets, including Level I by 2.616 bits.

The reduced-parameter implementation now composes the complete attack and produces publicly accepted forgeries from public inputs alone. Consequently, the result is no longer merely a collection of independently tested micro-lemmas. What remains conditional is the full-parameter complexity projection: the exact characteristic-polynomial degree transfer to the saturated dominant algebra, the fast online-product schedule, seeded-instance transfer, and memory traffic all require further scrutiny. The paper therefore claims a finite, executable structural attack and a model-based candidate full-parameter break, not a practical full-parameter computation.

The algebraic ceiling is exact, while the concrete bit-security estimates are conditional on the stated solver and cost model. The reduced-parameter implementation establishes that the attack components compose. Any QR-UOV revision should reconsider both dimensions and the underlying compressed structure rather than apply a vinegar-only retuning.

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